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Genetics& Genodermatoses

Genetic (inherited) skin diseases

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Lecture in Dermatology

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Genodermatoses are inherited genetic skin conditions often grouped into three categories: chromosomal, single gene, and polygenetic.

↳ multiple genes affected

Example of genodermatoses

- Epidermolysis bullosa
- Ichthyosis
- Palmoplantar keratoderma
- Neurofibromatosis
- Xeroderma pigmentosum
- Incontinentia pigmenti
- Restrictive dermopathy
- Pachyonychia congenita

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Epidermolysis Bullosa

bullae formation due to mechanical factor

*Is a group of mechanobullous disorders.

*There are over 20 different mutations associated.

*The main mutations effect keratin and collagen.

*Preventing proper anchoring of the skin and resulting in bullae formation.

with simple pressure and cleavage formation

*There are four major subgroups of EB, based on the location of the cleavage plane.

most superficial at epidermis level

at dermo-epidermal junction (Basement membrane)

below dermo-epidermal junction

*EB Simplex, Junctional EB, and Dystrophic EB with a new category called Kindler Syndrome

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Epidermolysis Bullosa

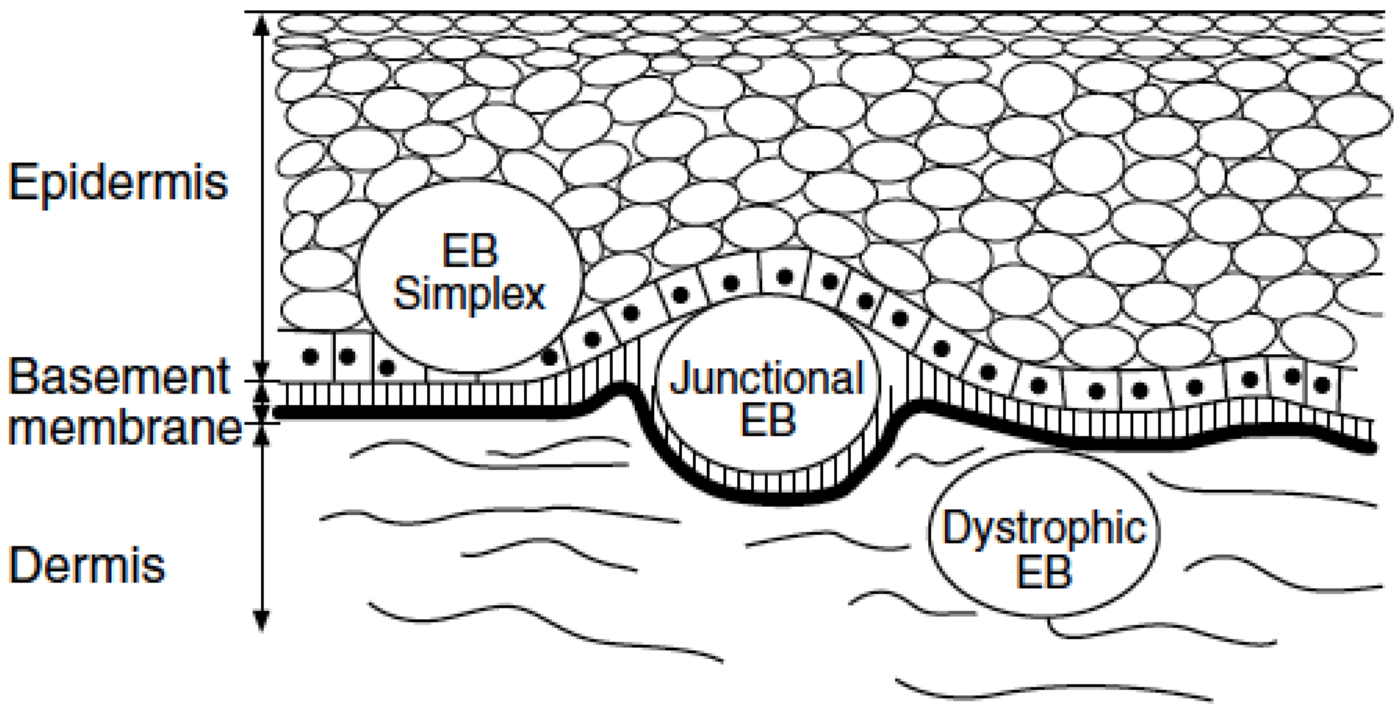

Figure 1
Level of cutaneous blisters in different types of epidermolysis bullosa. Adapted with permission (3).

Epidermolysis Simplex

Autosomal dominant → single defect allele cause disease if, no parent has disease and child gets it it's due to mutation

Prevalence: 1-2 in 100,000

Most Common overall

Mild disease

Affects epidermis superficial to the basement membrane

Blisters of then heal <u>without scarring</u>

→ since it only involve epidermis

Junctional Epidermolysis Bullosa

Severe autosomal recessive disorder

Mutation of the laminin 5 gene allowing separation between the dermis and epidermis

Death often before 2 years of age

Airway involvement

even death

Larynx affected—recurrent stridor and risk for asphyxiation

Recurrent oral lesions making feeding difficult

Sepsis

Poor nutritional state

Frequent severe blisters which can become colonized

Dystrophic Epidermolysis Bullosa

Most frequent type of EB seen by anesthesio logist

Prevalence: less than 1 in 100,000

Defect of the basement membrane and the

to mutations of collagen 7

Two forms:

Autosomal recessive (RDEB)- more common

Autosomal dominant (DDEB)

Defect of the basement membrane and the dermis due to mutations of collagen 7

Genetic Keratization Disorders:

These are group of different inherited abnormality in keratinization.

Disorder of keratinization is due to a defect in keratin metabolism.

In the normal epidermis, the process of terminal differentiation and cornification involves complex metabolic changes.

There are different syndromes, among which Ichthyosis group is the most prevalent one.

Ichthyosis

Ichthyosis is a disorder of cornification, characterised by persistently dry, thickened, 'fish scale' skin. → use ointment

↳ very dry very thick, mobility affected

There are at least 20 varieties of ichthyosis, including inherited and acquired forms.

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Inherited types of ichthyosis may be congenital or have delayed onset.

common → so in a disease with many clinical types the one with vulgaris is the most common

Ichthyosis vulgaris (1:250—1000) has an autosomal dominant inheritance, meaning an

→ affect male & female equally

abnormal gene is inherited from a parent. Penetrance is 90%. Onset is delayed until at least three months of age.

Recessive X-linked ichthyosis (1:2000—6000) mainly affects males, who have a single X chromosome with the abnormal gene. Females are protected by usually having a normal second X chromosome. Onset may be congenital or delayed by up to 6 months.

Treatment for ichthyosis

There is no cure for the inherited forms of ichthyosis..

Non-soap cleansers (soap may exacerbate dryness)

Bathing in salt water

Rubbing with a pumice stone or exfoliating sponge to remove scale

Moisturising creams containing urea, salicylic acid or alpha hydroxy acids. These are best applied to damp skin.

vitamin A analogue

In severe disease: oral retinoids acitretin and isotretinoin

Oral antibiotics for secondary bacterial infection

Vitamin D supplementation.

Neurofibromatosis

This relatively common disorder affects about 1 in 3000 people and is inherited as an autosomal dominant trait. There are two main types:

-Von Recklinghausen's neurofibromatosis (NF1; which accounts for 85% of all cases)

-Bilateral acoustic neurofibromatosis (NF2); these are phenotypically and genetically distinct.

In (NF1) :

Cafe

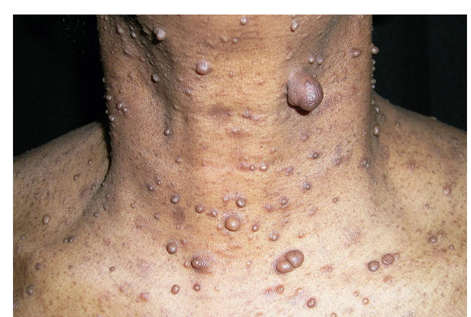
1.Six or more cafй au lait patches (light brown oval macules), usually developing in the first year of life.

freckles (dot dots)

2.Axillary freckling in two-thirds of affected individuals.

3.Variable numbers of skin neurofibromas, some small and superficial, others larger and deeper.

gradually increasing in number and size



Neurofibromas mostly are dome-like nodules. Some are firm, some soft and compressible through a deficient dermis (‘button-hole’sign); others feel ‘knotty’ or ‘wormy’.

Fibromas may not appear until puberty and become larger and more numerous with age.

4.Small circular pigmented hamartomas of the iris (Lisch nodules), appear in early childhood.

Nearly all NF1 patients meet the criteria for diagnosis by the age of 8 years, and all do so by 20 years.

The usual order of appearance of the clinical features is cafй au lait macules, axillary freckling, Lisch nodules and neurofibromas.

X not painful but disfigurin

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affecting hair follicle

KERATOSIS PILIARIS

Keratosis pilaris is a common disorder that is inherited as an autosomal dominant in childhood and reaches its peak incidence in adolescence.

This disorder leads to keratinous plugs in the follicular orifices with varying degrees of perifollicular erythema.

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Clinical Picture:

The lesions appear as small gray to white plugs of keratin that obstruct the mouths of the follicles entrapping the hair.

The sites of predilection are the extensor surfaces of the upper arms, thighs and buttocks. and sometimes cheeks

Some follicles are completely spared while adjacent ones are grossly plugged showing a (longitudinal) long strand of keratin surrounding the hair shaft protruding when examined in light (antenna sign).

Treatment of Keratosis Pilaris

Keratosis pilaris normally clears up on its own at a slow pace. In order to speed up recovery, treatment methods are available, which mainly focus on exfoliation and moisturizing the affected skin.

-Lactic acid lotions to reduce roughness and soften the tiny plugs

-Alpha hydroxy, or glycolic acid lotions to reduce scaling and help the skin retain moisture

-

Urea creams to moisturize and soften the skin and help loosen dead skin cells

-Salicylic acid lotion to soften and loosen dry, scaly or thickened skin

-Topical corticosteroids to reduce itching

-Topical retinoids, such as tretinoin, adapalene and tazarotene, to promote cell turnover and prevent hair follicles from plugging

It can take months to see any improvement and the tiny bumps almost always come back once stopping treatment.

Case Scenario:

Young lady aged 15yrs visited dermatology clinic for having skin lesions on the extensor surface of the arms and cheeks.

She mentioned that her mother and older brother have the same condition

On examination: multiple grey to white pinpoint papules on erythematous base was found.

What is the possible Dx? Keratosis Pillaris

What is the cause? Autosomal dominant)